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MoSDeF-dihedral-fit: A lightweight software for fitting dihedrals within MoSDeF

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Summary

Molecular Mechanics (MM) simulations (molecular dynamics and Monte Carlo) provide a third method of scientific discovery, simulation modeling, adding to the traditional theoretical and experimental scientific methods. These molecular simulations provide visualizations and calculated properties that are difficult, too expensive, or unattainable by conventional methods. Additionally, molecular simulations can be used to obtain insights and properties on chemicals or materials that do not currently exist, are not easily attainable, or require hard-to-achieve conditions (i.e., very high pressures and temperatures). However, these MM models require force field parameters determined from Quantum Mechanics (QM) simulations, where the MM proper dihedrals (i.e., dihedrals) are difficult to obtain if they don't currently exist for the chosen force field. While the same QM simulations can be used to fit the dihedrals in all of the force field types, these MM dihedrals are also not easily transferable between different force fields due to the differing parameters and formulas used in each force field (e.g., Combining rules and 1-4 scaling factors).

The `MoSDeF-Dihedral-Fit` (Crawford et al., 2023) library lets users quickly calculate the MM proper dihedrals (dihedrals) directly from the QM simulations for several force fields (OPLS, CHARMM, TraPPE, AMBER, Mie, and Exp6) (Brooks et al., 2009; Jorgensen et al., 1996; Lee et al., 2016; Martin & Siepmann, 1998; S. J. Weiner et al., 986 C.E.; Scott J. Weiner et al., 1984; **?**; **?**; **?**). The user simply has to generate or use an existing Molecular Simulation Design Framework (MoSDeF) force field XML file (Cummings et al., 2021; *Forcefield-Utilities*, 2022; *GMSO*, 2019; Summers et al., 2020), provide Gaussian 16 or Gaussian-style Quantum Mechanics (QM) simulation files that cover the dihedral rotation (typically, 0-360 degrees), and provide the molecular structure information in a mol2 format (Frisch et al., 2016). The `MoSDeF-Dihedral-Fit` software uses the QM and MM data to fit the dihedral for the specific force field, fitting the constants for the OPLS dihedral equation form with the correct combining rules and 1-4 scaling factors, as specified in the MoSDeF XML force field file. This software also accounts for multiple instances of the dihedral and the molecular symmetry in the molecule, and automatically removes all the unusable cosine power series values due to this symmetry. The user can set other dihedral energies in the molecule to zero, allowing for a more flexible and accurate dihedral fit; this allows the multiple dihedral's conformational energies to be calculated from a single dihedral angle, a strategy that was used in some of the original OPLS dihedral fits. The `MoSDeF-Dihedral-Fit` software analytically calculates the Ryckaert-Bellemans (RB)-torsions and the periodic dihedral from the OPLS dihedral. If another form of the dihedral equation that is not currently supported is needed, the software outputs the raw

data points to enable users to fit any other dihedral form. Therefore, the `MoSDeF-Dihedral-Fit` software allows the fitting of any dihedral form, provided the force fields and software it utilizes are supported by MoSDeF and MoSDeF-GOMC (which uses GPU Optimized Monte Carlo - GOMC) [Crawford:2023a; Crawford:2022; Crawford et al. (2023); Nejahi:2019; Nejahi:2021], vmd-python [vmd-python:2016] (a derivative or modified version of the original VMD software [Humphrey:1996; Stone:2001]), and the QM data is provided as a Gaussian output file, or a generalized Gaussian-style output form [Gaussian16:2016].

Statement of need

Many different types of Molecular Mechanics (MM) simulation models exist, which are also known as "force fields". While many of these force field parameters can be transferred between force fields, such as bonds, angles, and improper dihedrals (impropers), the proper dihedrals (dihedrals) can not be easily transferred due to the different combining rules (arithmetic and geometric) and 1-4 scaling factors (i.e., scaling factors between the 1st and 4th atoms) that were used in the development of the original parameters (Berthelot, 1898; Good & Hope, 1970; Lorentz, 1881). The accuracy of these dihedral parameters for each force field is critical for molecular simulations to obtain the correct molecular conformations and configurations, which are absolutely required for understanding and analyzing the system's microstructure and physical properties (e.g., free energies, viscosities, adsorption loading, diffusion constants, and many more).

While some dihedral fitting software currently exists, they only fit the CHARMM-style force fields (Mayne et al., 2013), or fit the dihedral constants to the final MM and QM energies, which need to be calculated by other means (Guvench & MacKerell, 1998). Therefore, the molecular simulation community needs a generalized software package that imports QM and MM files, automatically reads and organizes the QM data, calculates the MM energies, and automatically fits the dihedral. Additionally, the molecular simulation community needs software that fits the dihedral to any force field style and auto-corrects the fit to account for multiple instances of the dihedral and molecular symmetry, since fitting these dihedrals is a high barrier to simulating new chemistry and materials if these parameters do not exist for the chosen force field. The `MoSDeF-dihedral-fit` software accomplishes all this and automatically accounts for any of the common combining rules, allows the zeroing out any other dihedrals in the molecule (i.e., allowing a better fit and user control by setting the other dihedral energies to zero), and accounts for any 1-4 scaling factors specified via the MoSDeF XML files [Cummings:2021; Summers:2020; GMSO:2019; forcefield-utilities:2022], which contain the force fields. The `MoSDeF-dihedral-fit` (Crawford et al., 2023) API fills the missing gap by providing a generalized and easy solution to fitting dihedrals for any dihedral form that is allowable in the Molecular Simulation Design Framework (MoSDeF) and MoSDeF-GOMC (uses the GPU Optimized Monte Carlo - GOMC MM engine) software [Crawford:2023a; Crawford:2022; Crawford et al. (2023); Nejahi:2019; Nejahi:2021].

Commonly used force field dihedrals (Example: OPLS) were originally fit when the QM simulations used to fit the dihedrals were computationally prohibitive. Due to these limitations, scientists assumed that the dihedral fits were transferable with all the atom classes in the dihedral fit; however, this is not always an accurate assumption. Some of the dihedrals were only fit to the first minimum and not the entire dihedral landscape, which can lead to errors in the predicted molecular conformations. These prior assumptions in the dihedral fits may also lead to problems in reproducibility in modified force fields. Today, with more advanced hardware and software, QM simulations can be conducted with more complex molecules, allowing for higher quality and customized dihedral fits. The `MoSDeF-dihedral-fit` software will enable scientists to create a generalized, or molecule-specific dihedral parameters, quickly, accurately and reproducibly for any force field.

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Mathematics

Proper dihedral (dihedral) forms.

OPLS-dihedral:

$$OPLS_{Energy} = \frac{f_0}{2}$$

$$+\frac{f_1}{2} * (1 + cos(\theta)) + \frac{f_2}{2} * (1 - cos(2 * \theta))$$

$$+\frac{f_3}{2} * (1 + cos(3 * \theta)) + \frac{f_4}{2} * (1 - cos(4 * \theta))$$

Ryckaert-Bellemans (RB)-torsions:

$$RB_{Energy} = C_0$$

$$+C_1 * cos(\psi) + C_2 * cos(\psi)^2$$

$$+C_3 * cos(\psi)^3 + C_4 * cos(\psi)^4$$

$$\psi = \theta - 180^o$$

Periodic-dihedral:

$$Periodic_{Energy} = K_0 * (1 + cos(n_0 * \theta - 90^o))$$

$$+K_1 * (1 + cos(n_1 * \theta - 180^o)) + K_2 * (1 + cos(n_2 * \theta))$$

$$+K_3 * (1 + cos(n_3 * \theta - 180^o)) + K_4 * (1 + cos(n_4 * \theta))$$

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